
A Dublin core application profile for documenting analysis patterns in a reuse infrastructure

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Abstract: Analysis patterns are reusable computational artefacts aimed at the analysis stage of the software development process. Although analysis patterns can facilitate the work of analysts and developers, the access to them is still very poor because of the way they are usually described and made available. In order to reduce these limitations as well as to support the cataloguing and to encourage the reuse of analysis patterns, the Analysis Patterns Reuse Infrastructure (APRI) was proposed. This infrastructure comprises a repository of analysis patterns documented through a specific metadata profile that can be accessed via web services. Based on the proposal of APRI, this paper introduces a machine-processable metadata profile suited to the documentation of analysis patterns, called Dublin Core Application Profile for Analysis Patterns (DC2AP). This metadata profile is described by RDF files identified via URI, thus providing linked data that increase the potential for reusing the analysis patterns.

Keywords: analysis patterns; reuse; metadata standards; Dublin core; semantic web; linked data.

Reference to this paper should be made as follows: Vegi, L.F.M., Lisboa-Filho, J., Crompvoets, J., Soares, L.S. and Braga, J.L. (2013) 'A Dublin core application profile for documenting analysis patterns in a reuse infrastructure', *Int. J. Metadata, Semantics and Ontologies*, Vol. 8, No. 4, pp.267–281.

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This paper is a revised and expanded version of a paper entitled 'A machine-processable Dublin core application profile for analysis patterns to provide linked data, presented at 'International Conference on Dublin Core and Metadata Applications – DC'2012', Kuching, Malaysia, 3–7 September 2012.

1 Introduction

Analysis patterns are reusable computational artefacts, aimed at the reuse of ideas during the analysis stage of the development process of software. According to Fowler (1997), analysis patterns are ideas proven useful in a given practical context of software that may be useful and reused at the development of software for other practical contexts. Thus, these reusable computational artefacts can make the analysis stage faster and more accurate (Fernandez and Yuan, 2000), and help to avoid rework in the development and testing of solutions that already exist.

Although analysis patterns can facilitate the work of analysts and programmers by adding value through reuse of proven useful and tested ideas, the access to them is still very poor (Blaimer et al., 2010). So far there is no template to specify analysis patterns that are widely accepted, thus each set of analysis patterns is documented according to the preferences of their authors. In addition to not having a pattern specification, the analysis patterns are normally provided in scientific books and papers that are not machine-processable. These restrictions do not allow analysis patterns to be effectively retrieved by a search engine. In consequence to these restrictions, the time taken to retrieve an analysis pattern and the quality of the results obtained by searches are burdened (Blaimer et al., 2010). This happens because the understanding of the patterns documentations is restricted to human ability of interpretation.

In order to increase the potential for reuse of analysis patterns, the architecture of an Analysis Patterns Reuse Infrastructure (APRI) was proposed (Vegi et al., 2012). This architecture was inspired by the components of Spatial Data Infrastructures (SDI) proposed by Béjar et al. (2009), consisting of a repository of analysis patterns documented through a specific metadata profile that can be accessed via web services. APRI provides ways to minimise the problem of specification and retrieval of analysis patterns.

This paper presents an application profile of Dublin Core metadata standard (DCMI, 1998) created specifically for documenting analysis patterns in a manner consistent with the proposal of APRI (Vegi et al., 2012). The template proposed by Pantoquilha et al. (2003) provides a semantic base for the elements of the application profile as a way to describe analysis patterns. The application profile as presented in this work enables the documentation of analysis patterns as linked data (Bizer et al., 2009). This is possible through the use of Resource Description Framework (RDF) files (Klyne and Carroll, 2004), which is a machine-processable file format.

This paper is outlined as follows: Section 2 describes related work to the documentation of analysis patterns, the Dublin Core metadata standard, the semantic web and linked data. The machine-processable metadata profile proposed in this paper as the Dublin Core Application Profile for Analysis Patterns (DC2AP) is introduced in Section 3. Section 4

presents an example of an analysis pattern documented with the proposed metadata profile. Finally, Section 5 presents briefly the main conclusions and future work activities.

2 Related work

2.1 Documentation of analysis patterns

Documentation of analysis patterns is an important way for contextualising the reuse scope of a pattern and for enabling the sharing of knowledge among analysts and designers. However, this documentation happens in a heterogeneous manner among authors, since there is no standard to document analysis patterns (Blaimer et al., 2010).

There are many approaches to document analysis patterns, ranging from non-formalised textual descriptions to formalised descriptions based on templates.

Some examples of analysis patterns documented in a non-formalised textual manner can be found in the work of Fowler (1997) and Hay (1995). This manner impacts analysis patterns reuse, because it makes it hard for designers to understand the contextual scope of patterns, and it also limits the retrieval of analysis patterns through computerised search engines. Thus important detailed information for designers may not be documented or even retrieved, so limiting the spread of these patterns and thereby their potential for reuse.

Analysis patterns have also been described by the use of templates, which are structures with predetermined topics similar to those used to describe design patterns (Gamma et al., 1994). Usually a template is composed of essential topics such as context, problem, motivation and solution (Fowler, 1997), combined with other specific topics as defined by their authors. Some examples of analysis patterns that are documented through templates can be found in the works of Lisboa-Filho et al. (2002) and Fernandez and Yuan (2009). In the work of Lisboa-Filho et al. (2002), the analysis patterns presented are documented using a template as proposed by Meszaros and Doble

(1997). This template contains the following topics: name, problem, context, motivation, solution, participants and related patterns. Pantoquilha et al. (2003) detailed template that is developed specifically to document analysis patterns. This template combines common topics used previously by several authors, with new topics aimed to widen the documentation.

Another important factor for increasing the potential for reuse of analysis patterns is the way they are organised and available. Before a pattern is applied to a project, the designer needs to be aware of its existence in order to select (Blaimer et al., 2010). Usually, analysis patterns are organised together forming collections of patterns. A pattern collection is usually documented homogeneously by its author.

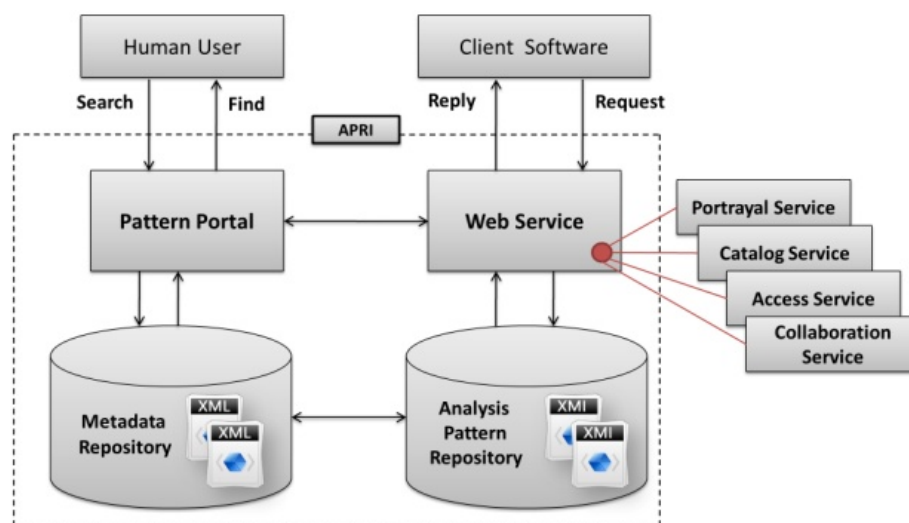
Collections of analysis patterns can be made available in different formats, such as books, articles and websites, and they can still be classified as pattern languages or pattern catalogues (Blaimer et al., 2010).

2.2 Analysis patterns reuse infrastructure

SDI is a relevant base collection of technologies, policies and institutional arrangements that facilitate the availability of and access to spatial data (Nebert, 2004). At this moment, most SDIs are based on Service-Oriented Architecture (SOA), allowing to create, shared, distributed and interoperable environments on the basis of web services (Davis-Júnior and Alves, 2005).

In the field of Geographic Information Systems (GIS), SDIs have been used very efficiently as a basic system for reusing services and geospatial data. In order to increase the potential for reuse of analysis patterns, an architecture for APRI was proposed, which is analogous to an SDI regarding the way data are documented, retrieved and published (Vegi et al., 2012). Similar to SDI, in the proposed infrastructure, the analysis patterns should be documented by making use of metadata (Duval, 2001) and can be retrieved through web services of various kinds. Figure 1 presents the proposed infrastructure architecture.

Figure 1 Architecture of the analysis patterns reuse infrastructure (Vegi et al., 2012) (see online version for colours)



APRI is composed of seven basic components: (a) *pattern portal* contains a set of tools and services focused on the retrieval, cataloguing and reuse of analysis patterns; (b) *metadata repository* contains metadata in XML/RDF format for documenting analysis patterns and services contained in APRI; (c) *analysis pattern repository* contains the solutions proposed by analysis patterns represented with diagrams in the XMI format; (d) *portrayal service* supports the visualisation of diagrams contained in analysis patterns of APRI; (e) *catalogue service* enables the cataloguing, retrieval and reuse of analysis patterns and services of APRI, based on their metadata; (f) *access service* allows to obtain analysis patterns; and (g) *collaboration service* allows analysts and designers to share their use experiences to improve analysis patterns in APRI.

2.3 Dublin core metadata standard

Metadata are not only data about data but also how they are commonly known. According to Sicilia (2006), metadata are a mechanism for expressing the 'semantics' of information, as a means to facilitate information understanding, searching and retrieval.

The metadata standards are collections of data elements that can be used to describe objects, thus providing interoperability degrees among data (Duval, 2001). According to the domain of the data to be documented, these description structures may have variations compared to other domains due to peculiarities of each domain. Thus, several metadata standards have already been proposed in order to meet the specific needs of some domains. An example of metadata standard for specific domain can be found in the work of ISO (2003).

The Dublin Core metadata standard (DCMI, 1998) appeared from a workshop held in Dublin city in the US state of Ohio. This event brought together professionals from several fields of knowledge in order to establish a generic metadata standard composed of a small set of recurring elements in all areas (NISO, 2001). This metadata standard has two levels: simple Dublin core and qualified Dublin core. The simple Dublin core is the most broadly known Dublin core level. This level consists of 15 elements applicable to document artefacts of different domains.

The qualified Dublin core has seven additional elements and provides qualifiers to further refine the meaning of a resource (DCMI, 2005). These qualifiers are classified into two categories: element refinement and encoding scheme. The element refinements are qualifiers which improve the specificity and precision of the metadata. They make the meaning of an element more specific and with a more restricted scope allowing a more detailed description of data. Thirty-three different qualifiers of this kind are provided by the qualified Dublin core.

Another category of Dublin core qualifiers is the encoding scheme. This qualifier basically represents rules for the encoding of values of an element. Encoding schemes include controlled vocabularies and formal notations to format a string. The association of encoding schemes with elements aims to standardise the filling of them and so can enhance the degree of semantic interoperability of metadata. A major advantage of the Dublin core standard is its

versatility. Although it is a very simple metadata standard and does not provide enough resources to describe data of complex domains (National Information Standards Organization, 2001), it can be specialised into metadata profiles for application in specific domains (Nogueras-Iso et al., 2005). Coyle and Baker (2009) described the steps for creating a Dublin core application profile.

2.4 Semantic web and linked data

The semantic web is seen as a layer of the web where it is possible to publish, obtain and use data that can be processed directly or indirectly by machines (Berners-Lee, 2000). The desire to extend the capabilities of conventional web, consisting of hypertext documents, for a global environment, where machine-processable structured data are widely published and used is not new. Berners-Lee et al. (1994) predicted that the web in which the objects were predominantly documents interpretable by humans would evolve into an environment with more machine-processable semantic information.

While the semantic web aims to transform the conventional web in an environment composed of machine-processable data, linked data provide the means to make this possible. Linked data are machine-processable structured data published on the web. These data have their well-defined meanings and are linked to other data of different sources through semantic relationships (Bizer et al., 2009). The main difference between conventional hypertext web and linked data is the types of relationships established among the data. While in conventional web the nature of the relationship between two documents is implicit and interpretable only by humans, in linked data the nature is explicit and also interpretable by computers (Bizer et al., 2009).

Linked data use three main technologies to support the semantic web: (a) Uniform Resource Identifiers (URIs) (Berners-Lee et al., 2005) are used to identify the data, (b) the Hypertext Transfer Protocol (HTTP) is used as a mechanism for data retrieval, and (c) Resource Description Framework (RDF) is used to describe the machine-processable data (Keizer et al., 2011).

RDF is a framework of machine-processable metadata used to describe both web documents and concepts of the real-world as a network, such as people and companies (Sauermaun et al., 2008). From the use of RDF to describe the data, it becomes possible to establish semantic relationships between them due to the way how data are encoded. Each relationship between RDF data consists of the triple formed by the concepts of subject, predicate and object, where subject and object are URI identifiers that represent the related data and predicate specifies the semantics of this relationship, i.e. the kind of relationship between the data (Klyne and Carroll, 2004).

The linked data build the so-called web of data (Bizer et al., 2009). The web of data created from the publication of linked data on the web enables the creation of new types of applications classified into three main categories: (a) linked data browsers, (b) linked data search engines, and (c) applications of specific domain. Linked data browsers allow users to navigate between data from different sources through semantic relationships established between them, and then to browse

through a particular source of data and from that to go through the entire network by relationships among the data. The linked data search engines allow the performance of complex searches that are similar to those offered by database management systems, thus allow data to be retrieved from web of data through searches that consider the nature of relationships among them (Bizer et al., 2009).

Linked data must be published on the web accompanied by metadata to enable potential users, humans or machines, to assess their quality and reliability (Hartig, 2009). These metadata can be provided by using the metadata standards, e.g. Dublin core (DCMI, 1998).

3 A Dublin core application profile for analysis patterns

In order to improve retrieval and reuse of analysis patterns by means of a documentation that allows a specific treatment for this domain carried out by a computer, this section describes the proposed application profile DC2AP. This profile is an extension of the Dublin core metadata standard for the domain of analysis patterns. The profile DC2AP is machine-processable and was developed based on a template proposed by Pantoquillo et al. (2003) for describing analysis patterns. The DC2AP provides detailed information about the analysis patterns which are not provided by the commonly used documentations for this kind of pattern. This detailed information is retrievable properly by search engines.

3.1 Semantic mapping between metadata elements

In contrast to the Dublin core metadata standard, which is generic and therefore aimed to document resources of several domains briefly, the template proposed by Pantoquillo et al. (2003) is designed specifically for the documentation of analysis patterns in detail. Due to such level of details for specific domain presented by the Pantoquillo et al.'s template, it was chosen to be used as a basis for the creation of DC2AP, complementing the Dublin core elements.

An important task for the creation of DC2AP was the realisation of a mapping between the elements proposed by the Dublin core and elements of the template proposed by Pantoquillo et al. (2003). In this mapping process, elements of both structures were compared and classified based on their semantic correspondences and some conceptual conflicts were identified. These conflicts are characterised by similar concepts that are expressed differently by each of the mapped structures. After identifying and resolving the existing conceptual conflicts between structures, it became possible to combine the elements of Dublin core and Pantoquillo et al.'s template, thus creating a single structure free of redundancies and semantic inconsistencies. Table 1 shows the result of the mapping between the Dublin core standard and Pantoquillo et al.'s template. Although Table 1 shows only the mapping between the elements of Simple Dublin Core and the Pantoquillo et al.'s template, the elements contained in qualified Dublin core were also considered in this comparative process, but none of them had equivalents in the template used.

Table 1 Mapping between Dublin core and Pantoquillo et al.'s template

<i>Simple Dublin Core</i>	<i>Pantoquillo et al.'s template</i>
Title	1 Name 2 Also Known As
Creator	3 History ^a
Subject	7 Context
Description	5 Problem
	6 Motivation
	7 Context
	8 Applicability 14 Examples ^a 18 Known Uses ^a
Publisher	No equivalent
Contributor	3 History ^a
Date	3 History ^a
Type	No equivalent
Format	No equivalent
Identifier	1. Name ^a
Source	15. Related Patterns ^a
Language	No equivalent
Relation	13 Anti-Patterns Trap ^a
	15 Related Patterns ^a
	16 Design Patterns ^a
Coverage	No equivalent
Rights	No equivalent
No equivalent	4 Structural adjustments
	9 Requirements
	9.1 Functional requirements
	9.2 Non-functional requirements
	9.3 Dependencies and contributions
	9.4 Conflict identification & guidance to resolution
	9.5 Priorities
	9.6 Participants
	10 Modelling
	10.1 Structure
	10.1.1 Class diagram
	10.1.2 Class description
	10.2 Behaviour
	10.2.1 Collaboration or sequence diagrams
	10.2.2 Activity diagrams
	10.2.3 State diagrams
	10.3 Solution Variants
	11 Resulting context
	12 Consequences
	17 Design guidelines

Note: ^a Partly equivalent.

Several mappings between elements of the Dublin Core and elements of other structures have already been performed and made available in the literature. An example of such mapping is presented in USLOC (2008). These mappings can improve both compared structures when the main objective is to create a way to document a specific artefact through structures used to another contexts.

3.2 Addition of new elements and application rules

From the mapping described in Table 1, all equivalent elements have been identified and merged, giving rise to the basic structure of the DC2AP. After this merge, most elements of the Pantoquilho et al.'s template that had direct equivalent mapping became element refinements of the others from the Dublin core. This happened because the Dublin core elements require specialisations that enable them to compose an application profile for a specific domain and the elements of Pantoquilho et al.'s template have the specialisation degree necessary to Dublin core.

During the merge process of the structures, which resulted in the set of elements that make up DC2AP, some elements underwent adaptations, new ones were proposed, as is indicated by the column 'New' in Table 3, and two elements from Pantoquilho et al.'s template were discarded. The element 'Applicability' was discarded because its semantics are very similar to the element 'Problem' and therefore could be considered as redundant. The element 'Structural adjustments' was also discarded to be unnecessary in the context of an application profile of metadata.

In order to control the application of DC2AP during the documentation of analysis patterns, rules on the obligatoriness, occurrence and type of value of each of the elements were proposed. These application rules and their acronyms are presented in Table 2.

Table 3 presents all the elements that make up the DC2AP associated with their respective application rules. Some of these elements are intended to version control and others are intended to share experiences of use. These features were incorporated into DC2AP to allow the creation of dynamic collections of analysis patterns, where newly improved versions of the patterns may be proposed from the experience of their usage. Moreover, all the versions of the analysis patterns may be related to each other, thereby providing the creation of a repository of analysis patterns that is rich of details about the patterns evolution. These resources allow potential users to retrieve the version that best meets their needs. All these characteristics are consistent with the proposal of the APRI (Vegi et al., 2012).

A brief description about the semantics of each of the elements that make up the DC2AP is presented in Appendix A. A complete description about the DC2AP elements, as well as some details about the application rules associated with them can be found in the work of Vegi (2012), where a detailed technical description of this metadata profile is presented.

Table 2 DC2AP application rules

<i>Rule Type</i>	<i>Rules and Acronyms</i>	<i>Description</i>
Obligation	Mandatory [M]	Element must always be filled in the documentation of an analysis patterns.
	Optional [O]	Element is optional, as it refers to additional information.
	Conditional [Cd]	Element must be filled if there is corresponding information.
Occurrence	Single [S]	Element can occur only once in the documentation of an analysis pattern.
	Multiple [Mu]	Element can occur multiple times in the documentation of an analysis pattern.
Value type	String [St]	Element should be filled with textual descriptions.
	Date [D]	Element must be filled with a date following a format previously specified.
	URI [U]	Element should be filled with a string used to identify a resource on the Internet.
	Null [N]	Element should not be filled with any value because their data are concentrated in its element refinements.
	URI, Number or String [UNS]	Element can be filled with a resource identifier, with a number or a textual description.
	URI or String [US]	Element can be filled with a resource identifier or a textual description.

3.3 Providing linked data with DC2AP

In order to transform the metadata profile previously presented in a machine-processable profile and therefore suit to the proposal of linked data, each of the elements proposed by DC2AP were associated with URI identifiers and described by RDF semantic properties. According to Coyle and Baker (2009), it is important before creating new RDF semantic properties to look for existing properties that are semantically equivalent to the elements proposed by an application profile. The use of known properties associated with elements of a profile increases the metadata semantic interoperability. Then some investigations were performed looking for element sets described as RDF that contains semantic properties equivalent to elements of DC2AP. The Open Metadata Registry tool (The Registry, 2005) was used to perform these searches. This tool provides open services to register and search element sets, metadata application profiles, vocabularies and syntaxes encoding schemes described as RDF. Table 4 presents the compatible element sets described as RDF selected after the searches. They were associated with elements of DC2AP.

Table 3 DC2AP element set

<i>Elements and their Application Rules</i>				<i>New</i>
1	Identifier [M] [S] [UNS]			
2	Title [M] [S] [St]	2.1	Alternative Title [O] [Mu] [St]	
3	Creator [M] [Mu] [St]			
4	Subject [M] [Mu] [St]			
		5.1	Problem [M] [S] [St]	
5	Description [M] [S] [N]	5.2	Motivation [M] [Mu] [St]	
		5.2.1	Example [M] [Mu] [St]	
		5.2.2	Known Uses [O] [Mu] [St] ^a	
		5.3	Context [M] [S] [St]	
6	Publisher [O] [Mu] [St]			
7	Contributor [Cd] [Mu] [St]			
8	Date [M] [S] [N]	8.1	Created [M] [S] [D]	
		8.2	Modified [Cd] [S] [D]	
9	Type [M] [S] [US]	9.1	Notation [M] [S] [St]	YES
10	Format [M] [Mu] [US]			
11	Source [Cd] [S] [UNS]			
12	Language [M] [S] [US]			
		13.1.	Is Version of [Cd] [S] [UNS]	
		13.2.	Is Replaced by [Cd] [Mu] [UNS] ^b	
		13.3.	Replaces [Cd] [Mu] [UNS] ^b	
		13.4.	Is Part of [O] [Mu] [UNS]	
13	Relation [Cd] [S] [N]	13.5.	Has Part [O] [Mu] [UNS]	
		13.6.	Is Designed With [O] [Mu] [UNS] ^a	YES
		13.7.	Should Avoid [O] [Mu] [UNS] ^a	YES
		13.8.	Complemented by [O] [Mu] [UNS] ^a	YES
		13.9.	About [Cd] [S] [St]	
14	Coverage [O] [Mu] [St]			
15	Rights [Cd] [Mu] [US]			
		16.1	Event Date [M] [S] [D]	YES
16	History [M] [Mu] [N] ^b	16.2	Author [M] [Mu] [St]	YES
		16.3	Reason [M] [S] [St]	YES
		16.4	Changes [Cd] [S] [St]	YES
		17.1	Functional Requirements [M] [Mu] [St]	
		17.2	Non-functional Requirements [O] [Mu] [St]	
17	Requirements [M] [S] [N]	17.3	Dependencies and Contributions [M] [S] [St]	
		17.3.1	Dependency Graph [M] [S] [U]	YES
		17.3.2	Contribution Graph [Cd] [S] [U]	YES
		17.4.	Conflict identification & Guidance to Resolution [Cd] [Mu] [St]	
		17.5	Priorities diagram [M] [S] [U]	
		17.6	Participants [M] [Mu] [St]	
		18.1.1	Use Case Diagram [M] [S] [U]	YES
		18.1.2	Collaboration/Sequence Diagrams [M] [Mu] [U]	
18	Modelling [M] [S] [N]	18.1.3	Activity/State Diagrams [O] [Mu] [U]	YES
		18.2.1	Class Diagram [M] [S] [U]	
		18.2.2	Class descriptions [M] [S] [U]	
		18.2.3	Relationship descriptions [M] [Mu] [St]	YES
		18.3	Solution Variants [O] [Mu] [U] ^a	
19	Resulting Context [O] [Mu] [St] ^a			
20	Design Guidelines [O] [Mu] [St] ^a			
21	Consequences [M] [S] [N]	21.1	Positive [M] [Mu] [St]	YES
		21.2	Negative [M] [Mu] [St]	YES

Notes: ^a Element to share use experiences; ^b Element to version control.

Table 4 Element sets selected for association

Name	URI
The Dublin Core Metadata Element Set, v1.1	http://purl.org/dc/elements/1.1/
Dublin Core Terms	http://purl.org/dc/terms/
RDA Group 1 Elements	http://rdvocab.info/Elements/
RDA Roles	http://rdvocab.info/roles/

The DC2AP elements refinements: *9.1 Notation*, *13.9 About* and *16.2 Author* were associated with RDF semantic properties found in RDA Group 1 elements and RDA roles. All other DC2AP elements and elements refinements arising from the Dublin core were associated with RDF semantic properties found in the Dublin Core Metadata Element Set, version 1.1 and Dublin core terms.

Besides associating known semantic properties to elements of DC2AP, it was also necessary to declare new semantic properties to describe the elements that were not associated with the sets found. All new RDF properties have been declared in accordance with the DCMI namespace policy (Powell et al., 2007), and then identified with URIs redirected by the server <http://purl.org> to RDF schema files on the server <http://dpi.ufv.br>. These new semantic properties together make up the element set called 'DC2AP Element Set'. This element set is identified by URI <http://purl.org/dc2ap/elements/>.

To avoid ambiguity between the terms used during the documentation of analysis patterns and improve the degree of semantic interoperability, some elements of DC2AP were also associated with vocabularies and syntaxes encoding schemes. Both vocabularies and syntaxes encoding schemes are used to standardise the metadata registration (Duval, 2001). Initially searches were performed by existing vocabularies and syntaxes encoding schemes that could standardise the filling of DC2AP elements. These searches were again made using the tool Open Metadata Registry (The Registry, 2005). In addition to standardising filling

some elements of DC2AP through association with vocabularies and syntaxes encoding schemes already existing, two new controlled vocabularies were proposed to fill gaps that could compromise the documentation of analysis patterns by means of ambiguous terms. Together, these two new controlled vocabularies proposed compose the DC2AP Vocabulary Encoding Schemes (DC2AP VES).

Although many DC2AP elements are associated with syntaxes encoding schemes which standardise a string format, some elements such as *16 History*, *17 Requirements* and *18 Modelling* that could also be associated with this kind of encoding schemes to organise their information were not. Instead of associating these elements with specific syntaxes encoding schemes, their semantics were divided among many element refinements. This decision was motivated to enable retrieval of knowledge based on filters focused on these element refinements. In this way, DC2AP allows stratified searches and also easier documentations for human understanding.

The vocabularies and syntaxes encoding schemes used in the definition of DC2AP are presented in Table 5, while the technical description of this metadata profile (Vegi, 2012) presents in detail the vocabularies and syntaxes encoding schemes associated with each element of it. Figure 2 shows the behaviour of the metadata repository of an APRI when analysis patterns are documented as linked data provided by DC2AP.

Due to text size limitations, the URIs referring to DC2AP elements, semantic properties are not presented in this paper, but they can be found in the work of Vegi (2012).

Dublin core metadata initiative provides four levels of metadata interoperability to classify specifications and applications that use its metadata standard (Nilsson et al., 2009). The metadata interoperability levels are built containing the previous levels, so Level 3 contains levels 1 and 2 specifications, and Level 4 specification conforms to the previous three.

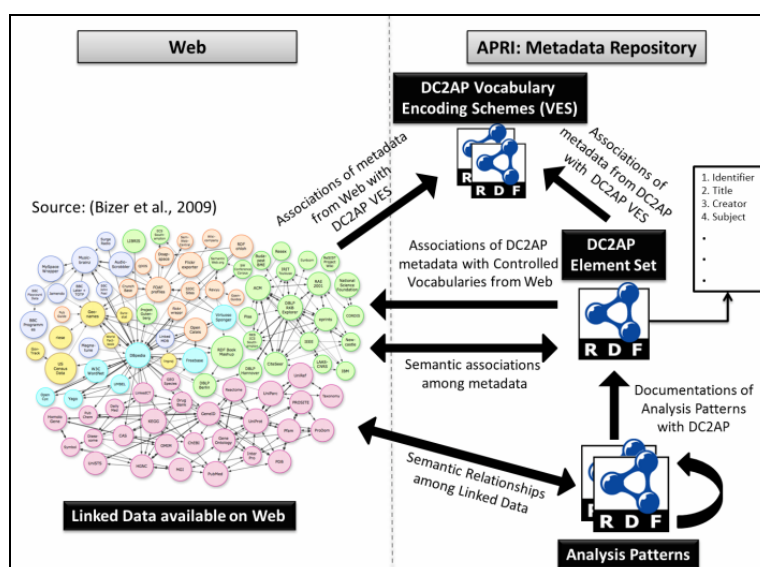
Figure 2 Providing linked data in an APRI (see online version for colours)

Table 5 Vocabularies and syntaxes encoding schemes associated to DC2AP

Name	URI	New
IMT	http://purl.org/dc/terms/IMT	
ISO 639-3	http://purl.org/dc/terms/ISO639-3	
Uniform Resource Identifier	http://purl.org/dc/terms/URI	
W3C-DTF	http://purl.org/dc/terms/W3CDTF	
DC2AP Type Vocabulary	http://purl.org/dc2ap/ves/type	YES
DC2AP Format Vocabulary	http://purl.org/dc2ap/ves/format	YES

The first interoperability level is named as informal interoperability. This level is applied to classify the Dublin core usages accompanied only by natural language definitions of the Dublin core elements. The second level corresponds to the formal semantic interoperability. At this level only applications are classified which implicitly or explicitly use the RDF semantics to define the Dublin core elements. The third level is about the description set syntactic interoperability. In order to be qualified at this level, it is necessary to informally map the Dublin core elements with rules about occurrence, obligation and value type to be expected. Finally, the fourth Dublin core interoperability level is applied to classify the usages of this metadata standard that provides a formal specification (in XML/RDF format for instance) to the rules to be expected on the third interoperability level.

Considering the four interoperability levels described above and the resources used during the DC2AP implementation, it is possible to mention that the DC2AP could be partly classified at the fourth interoperability level. The DC2AP element set is semantically described using RDF properties identified through URI. Besides this element set is

mapped with rules about occurrence, obligation and value type to be expected, and additionally, the major part of the DC2AP element set is associated with encoding schemes formally described by means of RDF triples.

A small metadata repository containing descriptions of analysis patterns made on RDF properties was created to exemplify the use of DC2AP in the documentation of analysis patterns interpretable by machines. This repository can be accessed from the URI <http://purl.org/dc2ap/Examples>. Peixoto et al. (2013) proposed a tool to document machine-processable analysis patterns providing linked data by means of RDF files. This tool, named as DC2AP Metadata Editor, allows users to create, edit and store metadata about analysis patterns according to elements and rules proposed in the DC2AP. The two major functionalities of DC2AP Metadata Editor are: metadata validation and generation of documentation in the RDF format.

The validation function verifies if the metadata obeys the DC2AP obligation rules, and the validated metadata can be exported to RDF files by means of the function to generate RDF. Both functionalities are also provided through a web service, enabling developers to reuse the DC2AP metadata editor resources using programming languages and technologies of their choice.

A beta version of DC2AP metadata editor can be found at: <http://purl.org/dc2ap/Editor>.

4 Example

In order to demonstrate the application of the metadata profile proposed in this work, Table 6 presents an example that uses the core of DC2AP to document the analysis pattern called urban street mesh, proposed by Lisboa-Filho et al. (2002). The core of a metadata profile corresponds to a subset of the profile elements that comprise the elements that are mandatory to be fill.

Table 6 Urban street mesh pattern documentation

Analysis pattern documented with DC2AP			
1	Identifier:	http://purl.org/apri/metadata/UrbanStreetMesh-v1	
2	Title:	Urban Street Mesh	
3	Creator:	Jugurta Lisboa-Filho, Cirano Iochpe and Karla A. V. Borges	
4	Subject:	Urban Management Systems, Geographic Information Systems and Streets	
5	Description	5.1	Problem: What are the elements that belong to the street mesh of a city?
		5.2	Motivation:
		–	Each drive way stretch is considered a road instance. It normally should be divided into several segments as well.
		–	A road stretch is a pathway segment between two connections.
		–	The set formed by the connections and road stretches create an urban street mesh.
		5.2.1	Example:
		–	Can be used as a basis for creating a system of urban bus routing.
			– Can be used as a basis for the creation of a system to calculate the shortest path between locations.
		5.3	Context: Almost every city in Brazil has shown the same organisation pattern, which is structured by their pathways organisation (e.g. streets, avenues, drives). The set of pathways stretches generates an urban street network.

Table 6 Urban street mesh pattern documentation (continued)

Analysis pattern documented with DC2AP		
6	Date	6.1 Created: 2002
7	Type: Analysis Pattern	7.1 Notation: UML
8	Format: JPEG and XMI	
9	Language: English	
10	History	10.1 Event Date: 2002
		10.2 Author: Jugurta Lisboa-Filho, Cirano Iochpe and Karla A. V. Borges
		10.3 Reason: Creation of this analysis pattern
		11.1 Functional Requirements:
		(R1) Register Crossroad - User must register all crossroads of an urban street mesh.
11	Requirements	(R2) Register Road Stretch - User must register all roads stretches of an urban street mesh.
		(R3) Register Road - User must register all roads of an urban street mesh.
		11.2 Dependencies and Contributions:
		R2 depends on R1 because every stretch begins at a crossroad and ends at another. R3 depends on R2 and R1 because every road is composed of roads stretches, and roads stretches have crossroads.
		11.2.1 Dependency Graph
12	Modelling	11.3 Priorities Diagram
		11.4 Participants: User.
		12.1.1 Use Case Diagram:
		12.1.2 Collaboration/Sequence Diagram
		12.1 Behaviour

Analysis pattern documented with DC2AP

6 Date 6.1 Created: 2002

7 Type: Analysis Pattern 7.1 Notation: UML

8 Format: JPEG and XMI

9 Language: English

10 History 10.1 Event Date: 2002

10.2 Author: Jugurta Lisboa-Filho, Cirano Iochpe and Karla A. V. Borges

10.3 Reason: Creation of this analysis pattern

11.1 Functional Requirements:

(R1) Register Crossroad - User must register all crossroads of an urban street mesh.

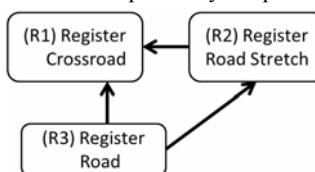
(R2) Register Road Stretch - User must register all roads stretches of an urban street mesh.

(R3) Register Road - User must register all roads of an urban street mesh.

11.2 Dependencies and Contributions:

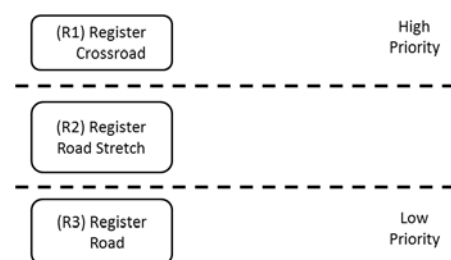
R2 depends on R1 because every stretch begins at a crossroad and ends at another. R3 depends on R2 and R1 because every road is composed of roads stretches, and roads stretches have crossroads.

11.2.1 Dependency Graph



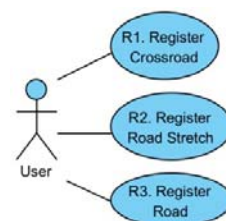
11 Requirements

11.3 Priorities Diagram

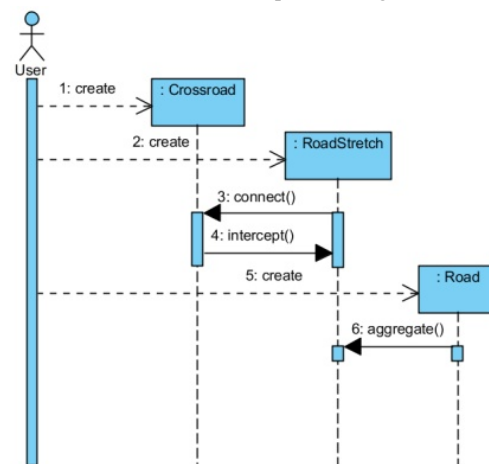


11.4 Participants: User.

12.1.1 Use Case Diagram:



12.1.2 Collaboration/Sequence Diagram



12 Modelling

12.1 Behaviour

Table 6 Urban street mesh pattern documentation (continued)

Analysis pattern documented with DC2AP		
12.2.1 Class Diagram		
<pre> classDiagram class StreetMesh { } class Road { id Name } class RoadStretch { idStretch initNum finalNum connect() aggregate() } class Crossroad { intercept() } StreetMesh "1" o-- "*" RoadStretch Road "1" o-- "*" RoadStretch RoadStretch "*" -- "2" Crossroad </pre>		
12.2.2 Class descriptions:		
12 Modelling	12.2 Structure	– Crossroad: This class represents the nodes of an street mesh that allow roads stretches are connected to each other. – Road Stretch: A road stretch can be connected to other stretches through instances of the class Crossroad. – Road: This is a conventional class usually implemented as a table in a relational DBMS. Each class is made of several roads stretches. – Street Mesh: This class is a geographic phenomenon represented by a complex spatial object. In this class can be defined many attributes relating to all mesh.
		12.2.3. Relationship descriptions:
		– The relationship between Road Stretch and Crossroad represents that each road stretch has a beginning and ending represented by a crossroad, being that a crossroad connecting various roads stretches. – The relationship between road and road stretch demonstrates that a road is an aggregate of several stretches.
13. Consequences		13.1. Positive: - This analysis pattern is very generic, so it fits in any context which involves urban street mesh, regardless of region or city size.
		- This pattern can easily be specialised for different contexts, thus having great potential for reuse.
		13.2. Negative: - Because it is a very generic analysis pattern, many projects can be served poorly for it, so is necessary specialise it to get better results in some cases.

5 Conclusions and future work

SDI is a widely accepted computational infrastructure focused strongly on the reuse of geospatial data and services. However the way in which SDI organise and promote the reuse of data can be very useful for the computing area in the reuse of various artefacts for software development, e.g. analysis patterns.

The use of analysis patterns speeds the software development process, since these computational artefacts provide generic solutions that can be reused as the initial basis for development. From the reuse of ideas provided by these patterns, common errors of analysis can be avoided, and the time required by analysts and developers during the conception of software can decrease. However, for a pattern to be reused, it is important that it is well-documented and made available in an accessible manner.

The Dublin core metadata standard was used in this study as a basis for creating a metadata profile to document analysis patterns (DC2AP) in an infrastructure specific to provide this reusable computational artefacts kind (APRI). Because of its capacity to be adapted for different contexts,

Dublin Core is shown to be adequate to documentation of patterns. The semantics of its elements can be specialised from the creation of element refinements and thus Dublin core can be used not only to adequately document the analysis patterns, but also for several other kinds of computational artefacts.

A machine-processable documentation is important to increase the potential for reuse of computational artefacts such as analysis patterns, so they can be retrieved more quickly and adequately by search engines. The elements and application rules of DC2AP were described as linked data to permit a better retrieval of analysis patterns by search engines. In this way, when analysis patterns are documented with DC2AP in an APRI, the metadata repository of this infrastructures becomes a source of specific domain linked data. This repository crosses the boundaries of APRI because it is linked to the global database composed of the linked data published on web.

As future work, it is intended to validate the acceptability of DC2AP through a quantitative research applied to the user community of analysis patterns, composed of system analysts, designers and developers. Besides, it is intended as well to use

the DC2AP as the basis for the definition and implementation of web services proposed in APRI. These services will be specific domain linked data applications and thus capable to facilitate the discovery of new sources of data from the RDF links existing in the analysis patterns documented in an APRI. This possibility of finding new sources of data will allow more up-to-dateness in metadata repositories of an APRI, since the services will be able to retrieve new related data found in other sources whenever they are published on the web. It is also intended to extend the APRI architecture to a more comprehensive and intelligent infrastructure, able to automatically recommend the reuse not only of analysis patterns, but also of other types of computational artefacts such as frameworks and design patterns. With these architecture extensions it will be possible to create new Dublin core application profiles to document the new reusable computational artefacts supported by APRI.

Acknowledgements

This work is financed by Brazilian funding agencies: FAPEMIG, CNPq and CAPES.

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Appendix A Dublin core application profile semantic description

- *Identifier*: An unambiguous reference given to an analysis pattern (AP) to differentiate it from others.
- *Title*: Formal name by which an AP is widely known.
- *Alternative title*: Alternative names used to substitute the formal name of an AP.
- *Creator*: Creator's name of an AP.
- *Subject*: Keywords related to the scope in which an AP is applied, thus providing a brief contextualisation.
- *Description*: Agglutinating element of descriptions aimed in the contextualisation of an AP over various aspects.
- *Problem*: Brief textual description of the problem solved by an AP.
- *Motivation*: Peculiarities of the problem which should be solved by the AP, regardless of application domain.
- *Examples*: Applications where the documented AP can be applied.
- *Known uses*: Real systems where an AP was applied.
- *Context*: Description of contextual coverage of the documented AP.
- *Publisher*: Names of responsible for providing the AP.
- *Contributor*: Names of responsible for changes made in the original version.
- *Date*: Agglutinating element of dates about the creation and modification of the documented AP.
- *Created*: Creation date of the version of the documented AP.
- *Modified*: Date of the last modification of the documented AP.
- *Type*: The nature of documented resource.
- *Notation*: The notation used to create the diagrams in the AP documentation.
- *Format*: Digital formats of diagrams used in documentation of an AP.
- *Source*: Reference to an AP used as the main source in the creation of an AP.
- *Language*: Language used for documenting an AP.
- *Relation*: Agglutinating element of existing relationships between the documented AP and other patterns.
- *Is Version Of*: Reference to the first version of an AP.
- *Is Replaced By*: References for all the most current versions of an AP.
- *Replaces*: References for all previous versions of an AP.
- *Is part of*: References for APs that contain the documented AP as part of its composition.
- *Has part*: References for APs which are contained as part of the documented AP.
- *Is designed with*: Known design patterns that can be used during the implementation of the documented AP.
- *Should avoid*: Anti-patterns that are important to avoid common errors identified during the application of the documented AP.
- *Complemented By*: Known APs that can be used to complement the AP.
- *About*: Description of details about relationships established among the documented AP and other patterns.
- *Coverage*: Spatial location or time period covered by the scope of an AP. It can compensate regionalisms that generate semantic ambiguities.
- *Rights*: Existing rights over the documented AP.
- *History*: Agglutinating element of historical data about the evolution of an AP.
- *Event date*: Date of the historic event occurrence.
- *Author*: Names of responsible for the occurrence of the historic event.
- *Reason*: Reason for the occurrence of the historical event.
- *Changes*: Changes made in the AP during an event of changes.
- *Requirements*: Agglutinating element of data about the requirements of an AP.
- *Functional requirements*: Responsibilities and restrictions for meeting each functional requirement of an AP.
- *Non-functional requirements*: Responsibilities and restrictions for meeting each non-functional requirement of an AP.
- *Dependencies and contributions*: Brief explanation about the identified relations of dependency and contributions among the requirements of an AP.
- *Dependency graph*: Graph diagram used to represent the identified relations of dependency among the requirements of an AP.
- *Contribution graph*: Graph diagram used to represent the identified relations of contribution among the requirements of an AP.
- *Conflict identification & guidance to resolution*: Brief explanation about the identified conflicts among the requirements of an AP and manners to solve them.

- *Priorities Diagram*: It represents the order which AP requirements should be satisfied respecting their dependencies.
- *Participants*: Entities that interact with AP requirements.
- *Modelling*: Agglutinating element of diagrams and descriptions that represents part of the problem solution.
- *Behaviour*: Agglutinating element of diagrams that make up the behavioural model of an AP.
- *Use case diagram*: Is used to represent a system that meets the requirements of an AP together with entities which interact with them.
- *Collaboration/sequence diagrams*: Are used to create scenarios that represent the execution of each use case of an AP.
- *Activity/state diagrams*: Are used to represent the general behaviour of the system specified by an AP.
- *Structure*: Agglutinating element of the diagram and descriptions that make up the structural model of an AP.
- *Class diagram*: Is the main part of the problem solution treated by an AP.
- *Class description*: Brief description of each class of the class diagram.
- *Relationship descriptions*: Brief description of the main relationships presented in the class diagram.
- *Solution variants*: Alternative models of behaviour and structure to solve the problem treated by an AP.
- *Resulting context*: Adaptations for an AP when it is applied in specific systems.
- *Design guidelines*: General tips to implement an AP.
- *Consequences*: Agglutinating element of descriptions of advantages and disadvantages of use the documented AP.
- *Positive*: Positive consequences of using the documented AP.
- *Negative*: Negative consequences of using the documented AP.